

Heating Curve

What a flat stretch on a heating curve is actually costing, and why boiling is the expensive part of the trip.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/heating-curve/

1 Before you open anything

Answer from what you already know. Nothing is being checked yet — the point is to have something on paper that the demonstration can later settle.

PREDICTION

A 100 W hot plate is heating 20 g of ice. It is never switched off: 100 J arrives every second, all the way through. Follow the stretch that begins when the first drop of liquid appears and ends when the last of the ice is gone. Over that stretch, what happens to (a) the thermometer reading and (b) the average speed of the particles?

In one sentence, say what reasoning led you to that.

2 Reading the whole trip once

Open the demonstration. The curve runs from $-25\text{ }^{\circ}\text{C}$ ice to $125\text{ }^{\circ}\text{C}$ steam. The **Right now** panel reports the single point you are stopped at; the **Where the energy goes** panel reports the five steps of the whole trip.

1. Set **Mass of water** to **20 g**. The value beside the label should read exactly that.
2. Press **Reset**. Fill in the first row of Table A from the four boxes under **Right now**.
3. Press **Heat it** and let it run all the way to the right-hand end (about 40 seconds). Fill in the second row of Table A.
4. Copy the **Energy** and **Share** columns of the **Where the energy goes** table into Table B.

Table A — the two ends of the curve

Where you are	Temperature	State	Total in	This step
After Reset				
Far right end				

Table B — where the energy goes, for 20 g

Step	Energy	Share
Warm ice $-25 \rightarrow 0\text{ }^{\circ}\text{C}$		
Melt at $0\text{ }^{\circ}\text{C}$		
Warm water $0 \rightarrow 100\text{ }^{\circ}\text{C}$		
Boil at $100\text{ }^{\circ}\text{C}$		
Warm steam $100 \rightarrow 125\text{ }^{\circ}\text{C}$		

a. Which single step costs the most, and what share of the total is it? Which step costs the least?

b. Add the Share values of the two steps that happen at a fixed temperature. Compare your sum with the figure the demonstration prints under the bar (...% **broke attractions**). Write both numbers.

c. At the far right end, what does **Time on a 100 W heater** read? _____

The demonstration states an assumption about that heater directly beneath the reading. Write it down, and say which way the real time would differ.

3 Crossing a flat part on purpose

Keep the mass at 20 g. You can drag the dot directly on the curve, or use the **Energy added** slider with the arrow keys for finer control. This part is the one that decides whether the graph makes sense to you.

1. Move the dot into the widest flat band — **State** reads **Water + steam, boiling**.
2. Stop five times as you cross it: just after it starts, about a quarter across, about halfway, about three-quarters across, and just before the curve tilts upward again.

3. At each stop record all four readings. **Average particle speed** is under the particle box.

Stop	Total in	This step	Temperature	Average particle speed
Just started				
About $\frac{1}{4}$ across				
About halfway				
About $\frac{3}{4}$ across				
Just before the end				

Now do the same on a sloped stretch. Move the dot back until **State** reads **Liquid water** and stop three times: near the low end, near the middle, and near the top.

Stop	Total in	Temperature	Average particle speed
Low end			
Middle			
Near 100 °C			

a. In the first table, which columns changed as you crossed and which did not? Answer the same question for the second table.

b. The particle picture changes completely across the flat part while one of your columns stays fixed. Energy is still arriving the whole time. Where is it going?

c. State the rule in your own words: during a phase change, energy added does _____ while temperature does _____, because the energy is doing _____ rather than _____.

d. Go back to your prediction in part 1. Which half of it did the demonstration confirm, and which half — if any — did it contradict? Name the specific assumption behind anything that turned out not to hold.

The black tick on the speed meter marks the speed at the *start* of the step you are currently in. Watch where the bar sits relative to that tick on a flat part, then on a sloped part, and say in one line what the tick is telling you.

4 Using it the way a question would

A. A shortcut, and where it breaks. A reasonable first move on “heat 20 g of water from $-25\text{ }^{\circ}\text{C}$ to $125\text{ }^{\circ}\text{C}$ ” is one line of $q = mc\Delta T$. Try it and see what it costs you.

1. On paper, calculate $q = mc\Delta T$ using $m = 20\text{ g}$, $c = 4.18\text{ J/g}\cdot^{\circ}\text{C}$ and $\Delta T = 150\text{ }^{\circ}\text{C}$. Show the arithmetic.
2. With the mass still at 20 g, drag to the far right end and read **Total in**.

Shortcut answer: _____ Demonstration **Total in**: _____

a. By what factor does the shortcut fall short? Name every term it left out, and state what quantity each of those terms needs instead of c and ΔT .

b. The shortcut also used one c for the whole range. Look at the **Constants used** panel and say why that is wrong even for the sloped parts alone.

B. Working backwards. A test question rarely hands you the mass.

1. Use your 20 g total from part 2 to work out the energy the whole trip costs *per gram*.
2. From that, calculate the mass whose whole trip would cost close to 100 kJ. Round to the nearest whole gram.
3. Set **Mass of water** to that value, drag to the far right end, and record what the demonstration gives.

Energy per gram: _____ Mass predicted: _____

Demonstration **Total in**: _____ **Time on a 100 W heater**: _____

c. State in one sentence how total energy depends on mass, and say why that let you scale a single number instead of redoing five separate calculations.

5 Solve these without the demonstration

Close the page or look away from it. Show your working. Then reopen it, run the check written under each problem, and if you disagree with the demonstration write down where your working went off.

Constants, as listed in the demonstration, at 1 atm: $c_{\text{ice}} = 2.09 \text{ J/g}^\circ\text{C}$ · $c_{\text{water}} = 4.18 \text{ J/g}^\circ\text{C}$ · $c_{\text{steam}} = 2.01 \text{ J/g}^\circ\text{C}$ · $\Delta H_{\text{fus}} = 334 \text{ J/g}$ at 0°C · $\Delta H_{\text{vap}} = 2260 \text{ J/g}$ at 100°C

1. How much energy does it take to bring 45.0 g of ice from -25.0°C to liquid water at 0.0°C ? Give your answer in kJ.

Check: set the mass to 45 g and drag until **State** has just changed from **Ice + water, melting** to **Liquid water**. Read **Total in**.

2. A 100 W heater delivers all of its energy to 20.0 g of liquid water that is already at exactly 100.0°C . How long until the last of it has become steam? Give the answer in seconds and in minutes.

Check: at 20 g, read **Time on a 100 W heater** at the start of the boiling plateau and again at its end, and subtract.

3. Find the total energy needed to take 15.0 g of liquid water at 20.0°C to steam at 110.0°C . Write each term separately before adding.

Check: at 15 g, record **Total in** with the dot as close as you can get to 20.0°C liquid water, then again as close as you can get to 110.0°C steam, and subtract. Landing exactly on those temperatures is not possible, so expect agreement to a few tenths of a kJ, not better.

4. Two identical 100 W plates. Plate A holds 20 g of ice sitting at exactly 0 °C; plate B holds 20 g of liquid water sitting at exactly 100 °C. Both samples change phase completely. Which finishes first, and by what factor? Then explain the difference in terms of what the energy has to do to the attractions between particles in each case — not by quoting the two constants.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet used.

- While you are on the first flat part, the equation line under **Right now** reports how many grams have melted so far. Stop where exactly half the sample has melted and check **This step** against your own $(m/2)(\Delta H_{\text{fus}})$.
- Set **Mass of water** to 1 g, then to 100 g. Does any figure in the **Share** column move? What does that tell you about what the shape of the curve depends on, and what it does not?
- The **Constants used** panel says both ΔH values come from the molar figures 6.01 kJ/mol and 40.7 kJ/mol divided by 18.02 g/mol. Do those two divisions yourself and see how closely they land on 334 and 2260 J/g.
- The caption under the particle box says a real gas is roughly a thousand times less dense than the liquid, and that the box cannot show it honestly. Work out roughly how much wider the box would have to be to draw the gas at true relative spacing, and decide whether the picture would still be worth looking at.